



National  
Qualifications  
2019

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**2019 Music Technology**  
**Higher**  
**Finalised Marking Instructions**

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## General marking principles for Higher Music Technology

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

- (a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- (b) You should not accept any answer that is not in the marking instructions. Use professional judgement when candidates' responses do not exactly match the marking instructions but carry the same meaning.
- (c) Accept inaccurate spelling of English or Italian terms, as long as the meaning is clear.
- (d) Where a question asks for a description, explanation or justification, do not award marks for simple one or two word answers.

## Marking instructions for each question

| Question |     |      | Expected answer(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Max mark                   | Additional guidance                                                                                                                                                                                |
|----------|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.       | (a) |      | Reggae<br>Counter melody.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2                          | 1 mark for each.<br><br>Do not accept other answers here.                                                                                                                                          |
|          | (b) |      | <ul style="list-style-type: none"> <li>beats 2 and 4 emphasised</li> <li>repeated riff pattern</li> <li>off-beat chords</li> <li>simple repeated chord pattern</li> <li>expressive and soulful vocals</li> <li>relaxed performance</li> <li>backing vocals in harmony</li> <li>prominent bass part</li> <li>syncopated bassline.</li> </ul>                                                                                                                                                                                                                                                                                                    | 2                          | 1 mark for each.<br><br>Do not accept other answers here.                                                                                                                                          |
| 2.       | (a) |      | <p>Jazz funk</p> <p><b>Possible music features</b></p> <ul style="list-style-type: none"> <li>ostinato and/or riff</li> <li>syncopation</li> <li>extended groove sections</li> <li>melodic and/or complex bass lines</li> <li>jazz instrumentation with electric instruments</li> <li>use of synthesiser</li> <li>use of Hammond organ</li> <li>prominent bass.</li> </ul> <p><b>Possible technological features</b></p> <ul style="list-style-type: none"> <li>live sounding recording with minimal post-production</li> <li>use of modulation effects</li> <li>use of synthesiser</li> <li>use of Hammond organ</li> <li>wah wah.</li> </ul> | <p>1</p> <p>1</p> <p>1</p> | <p>1 mark for correct genre.</p> <p>1 mark for a reason related to a musical concept.</p> <p>1 mark for a reason related to a technological concept.<br/><br/>Also accept chorus/flange/phase.</p> |
|          | (b) |      | Cross rhythms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                          | Do not accept other answers here.                                                                                                                                                                  |
| 3.       | (a) | (i)  | <p><b>Fault</b></p> <ul style="list-style-type: none"> <li>there is popping and blasting present in the mix</li> <li>mic technique/placement is poor.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                          | Do not accept other answers here.                                                                                                                                                                  |
|          |     | (ii) | <p><b>Correction</b></p> <ul style="list-style-type: none"> <li>adjust the mic position</li> <li>move the mic off axis</li> <li>use a pop shield.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1                          | Do not accept other answers here.                                                                                                                                                                  |



| Question       |                                                                                                                                                                                    | Expected answer(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Max mark       | Additional guidance                                                                                     |               |                                                                                                                          |                |                                                                                                                                      |                |                                                                                                                                         |               |                                                                                                                                                                                    |                |                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.             | (a)                                                                                                                                                                                | <p>Examples of key innovators, attributes and technologies <b>could</b> include:</p> <p>Genre: Hip hop</p> <table><tr><td>Key innovators</td><td><ul style="list-style-type: none"><li>DJ Kool Herc</li><li>Dr Dre</li><li>Grand Master Flash.</li></ul></td></tr><tr><td>Key attribute</td><td><ul style="list-style-type: none"><li>mixing using DJ decks</li><li>rapping/mc'ing</li><li>triggering samples.</li></ul></td></tr><tr><td>Key technology</td><td><ul style="list-style-type: none"><li>DJ decks</li><li>drum machine</li><li>sampler</li><li>sequencer</li><li>synthesiser.</li></ul></td></tr></table> <p>Genre: Jazz funk</p> <table><tr><td>Key innovators</td><td><ul style="list-style-type: none"><li>Herbie Hancock</li><li>Lonnie Liston Smith</li><li>Snarky Puppy</li><li>Stanley Clarke.</li></ul></td></tr><tr><td>Key attribute</td><td><ul style="list-style-type: none"><li>electric/electronic instrumentation with improvisation</li><li>irregular time signatures</li><li>syncopation</li><li>time changes.</li></ul></td></tr><tr><td>Key technology</td><td><ul style="list-style-type: none"><li>modulation/filter effects on guitar/wah wah</li><li>rhodes piano</li><li>synthesiser.</li></ul></td></tr></table> | Key innovators | <ul style="list-style-type: none"><li>DJ Kool Herc</li><li>Dr Dre</li><li>Grand Master Flash.</li></ul> | Key attribute | <ul style="list-style-type: none"><li>mixing using DJ decks</li><li>rapping/mc'ing</li><li>triggering samples.</li></ul> | Key technology | <ul style="list-style-type: none"><li>DJ decks</li><li>drum machine</li><li>sampler</li><li>sequencer</li><li>synthesiser.</li></ul> | Key innovators | <ul style="list-style-type: none"><li>Herbie Hancock</li><li>Lonnie Liston Smith</li><li>Snarky Puppy</li><li>Stanley Clarke.</li></ul> | Key attribute | <ul style="list-style-type: none"><li>electric/electronic instrumentation with improvisation</li><li>irregular time signatures</li><li>syncopation</li><li>time changes.</li></ul> | Key technology | <ul style="list-style-type: none"><li>modulation/filter effects on guitar/wah wah</li><li>rhodes piano</li><li>synthesiser.</li></ul> | 3 | <p>Candidates are asked to select a genre from the list provided. No marks are awarded for this.</p> <p>For the genre selected marks are awarded for identifying</p> <ul style="list-style-type: none"><li>a key innovator (1 mark)</li><li>a key attribute (1 mark)</li><li>a key technology (1 mark)</li></ul> |
| Key innovators | <ul style="list-style-type: none"><li>DJ Kool Herc</li><li>Dr Dre</li><li>Grand Master Flash.</li></ul>                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |                                                                                                         |               |                                                                                                                          |                |                                                                                                                                      |                |                                                                                                                                         |               |                                                                                                                                                                                    |                |                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                  |
| Key attribute  | <ul style="list-style-type: none"><li>mixing using DJ decks</li><li>rapping/mc'ing</li><li>triggering samples.</li></ul>                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |                                                                                                         |               |                                                                                                                          |                |                                                                                                                                      |                |                                                                                                                                         |               |                                                                                                                                                                                    |                |                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                  |
| Key technology | <ul style="list-style-type: none"><li>DJ decks</li><li>drum machine</li><li>sampler</li><li>sequencer</li><li>synthesiser.</li></ul>                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |                                                                                                         |               |                                                                                                                          |                |                                                                                                                                      |                |                                                                                                                                         |               |                                                                                                                                                                                    |                |                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                  |
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| Question              |                                                                                                                                                                                                                                                                                                                                                     |  | Expected answer(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Max mark              | Additional guidance                                                                                              |                      |                                                                                                                                                                                                                                                                                                                                                     |                       |                                                                                                                 |  |  |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------|--|--|
|                       |                                                                                                                                                                                                                                                                                                                                                     |  | <p>Genre: Reggae</p> <table><tr><td><b>Key innovators</b></td><td><ul style="list-style-type: none"><li>• Bob Marley</li><li>• King Tubby</li><li>• Lee ‘Scratch’ Perry.</li></ul></td></tr><tr><td><b>Key attribute</b></td><td><ul style="list-style-type: none"><li>• beats 2 and 4 emphasised</li><li>• repeated riff pattern</li><li>• off-beat chords</li><li>• simple repeated chord pattern</li><li>• expressive and soulful vocals</li><li>• relaxed performance</li><li>• backing vocals in harmony</li><li>• prominent bass part</li><li>• syncopated bassline.</li></ul></td></tr><tr><td><b>Key technology</b></td><td><ul style="list-style-type: none"><li>• Hammond organ</li><li>• spring reverb</li><li>• (tape) delay.</li></ul></td></tr></table> | <b>Key innovators</b> | <ul style="list-style-type: none"><li>• Bob Marley</li><li>• King Tubby</li><li>• Lee ‘Scratch’ Perry.</li></ul> | <b>Key attribute</b> | <ul style="list-style-type: none"><li>• beats 2 and 4 emphasised</li><li>• repeated riff pattern</li><li>• off-beat chords</li><li>• simple repeated chord pattern</li><li>• expressive and soulful vocals</li><li>• relaxed performance</li><li>• backing vocals in harmony</li><li>• prominent bass part</li><li>• syncopated bassline.</li></ul> | <b>Key technology</b> | <ul style="list-style-type: none"><li>• Hammond organ</li><li>• spring reverb</li><li>• (tape) delay.</li></ul> |  |  |
| <b>Key innovators</b> | <ul style="list-style-type: none"><li>• Bob Marley</li><li>• King Tubby</li><li>• Lee ‘Scratch’ Perry.</li></ul>                                                                                                                                                                                                                                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                                                                                                                  |                      |                                                                                                                                                                                                                                                                                                                                                     |                       |                                                                                                                 |  |  |
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| Question |     | Expected answer(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                   |
|----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | (b) | <p>Examples could include</p> <p><b>Wax Cylinder</b></p> <p><b>Example of key innovator</b><br/>Thomas Edison.</p> <p><b>Example of their influence on the development of this technology</b><br/>the wax cylinder was one of the first commercial mediums. Edison perfected a solid wax cylinder which could be shaved down and used again.</p> <p><b>Synthesiser</b></p> <p><b>Example of key innovator</b><br/>Bob Moog.</p> <p><b>Example of their influence on the development of this technology</b><br/>Moog developed a voltage controlled series of modules which could be patched together to create sound.</p> <p><b>Microphone</b></p> <p><b>Example of key innovator</b><br/>Dr Harry F Olson.</p> <p><b>Example of their influence on the development of this technology</b><br/>Olson was a key innovator of the ribbon microphone. This bi-directional microphone was used extensively in the broadcasting industry.</p> | 2        | <p>Candidates are asked to select a technology from the list provided. No marks are awarded for this.</p> <ul style="list-style-type: none"> <li>• Example of a key innovator<br/>(1 mark)</li> <li>• Example of the key innovator's influence on the development of the selected technology.<br/>(1 mark)</li> </ul> |

| Question |  |  | Expected answer(s)                                                                                                                                                                                                                                              | Max mark | Additional guidance                                                                                                                                                                                                              |
|----------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.       |  |  | 1. parametric<br>2. inverted pedal<br><br>3. portamento and phaser<br><br>4. release<br>5. submix<br>6. rallentando                                                                                                                                             | 7        | 1 mark for each correct answer.<br>1 – Also accept notch or bell curve.<br>3 – Also accept flanger<br>Also accept glide.<br>4 – Also accept sustain.<br>5 – Also accept group or bus.<br>6 – Also accept rall or ritardando/rit. |
| 9.       |  |  | The correct 5 features are <ul style="list-style-type: none"> <li>• arpeggios/broken chords on electric guitar</li> <li>• oboe plays a solo</li> <li>• tambourine panned right</li> <li>• time changes</li> <li>• tremolo effect on electric guitar.</li> </ul> | 5        | 1 mark for each correct feature.<br><br>Features can be listed in any order.<br><br>Do not accept other answers here.                                                                                                            |

[END OF MARKING INSTRUCTIONS]